LAB 3 REPORT

Course: Software Engineering
Lab: Lab 3

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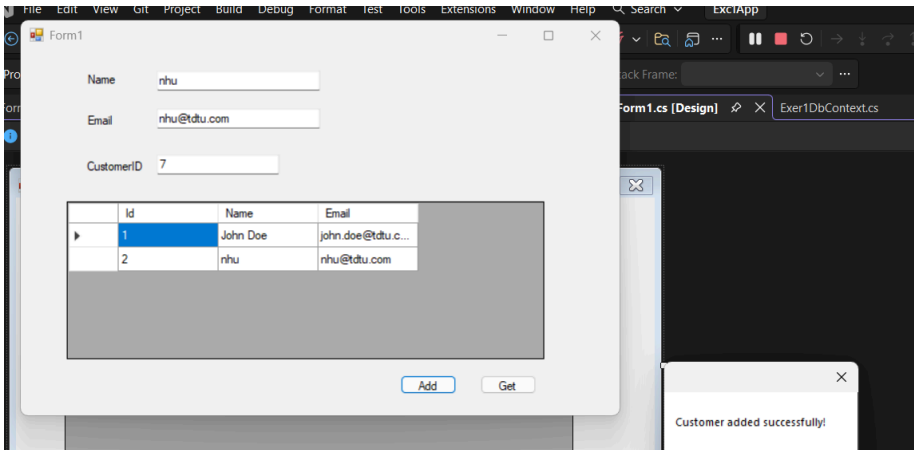
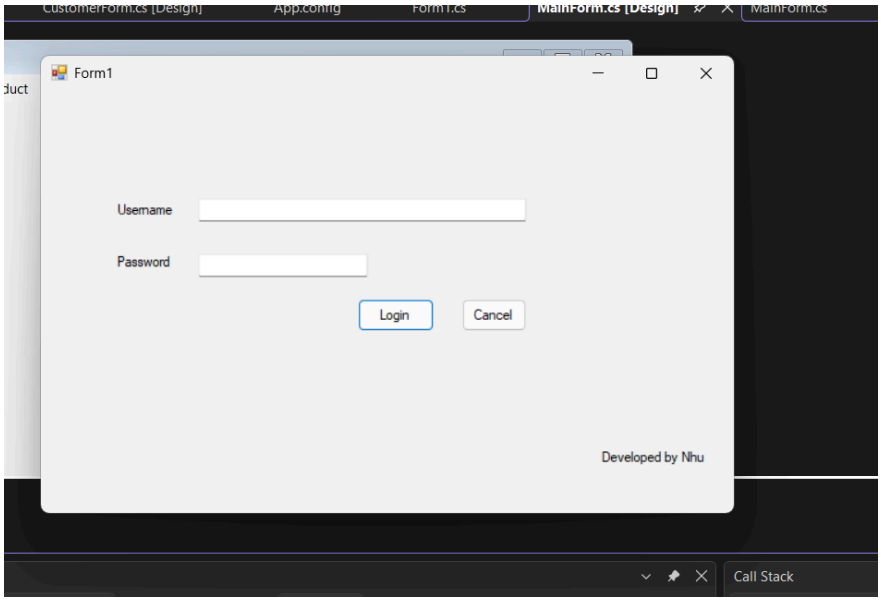
I. Introduction

- The purpose of this lab is to help students understand how to separate the application into three independent layers: Presentation, Business Logic, and Data Access.
- 3-Tier Architecture divides software into distinct layers to improve maintainability, scalability, and code organization. This design ensures each layer has a single responsibility and communicates with others through defined interfaces.

- Exercise:

I created the exc1 database, the Customers table, configured the connection string, and wrote the classes Customer, CustomerRepository, and CustomerService. After that, I designed a Form interface that allows entering Name and Email and displays the customer list using a DataGridView. I also fixed issues that occurred, such as Entity Framework connection errors, missing DLL build errors, and the error where the table already existed in SQL Server. After fixing them, the application ran successfully, displayed data, and added new customers to the database. I combined Exercises 2 and 3 together. I created the login feature from Lab 2 and organized it using the 3-tier architecture. To save time, I reused and connected to the existing SQL database from Lab 2 instead of creating a new one.




Form1
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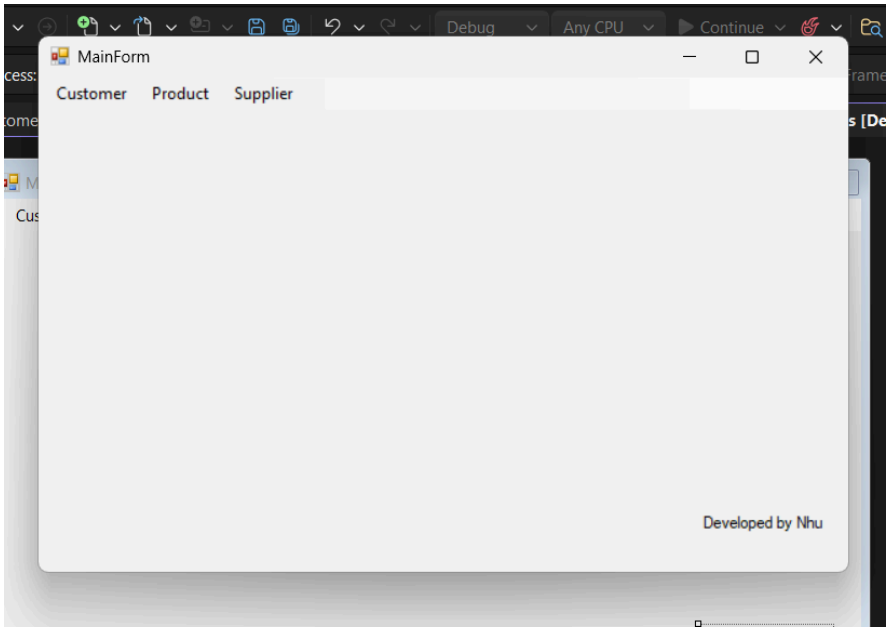
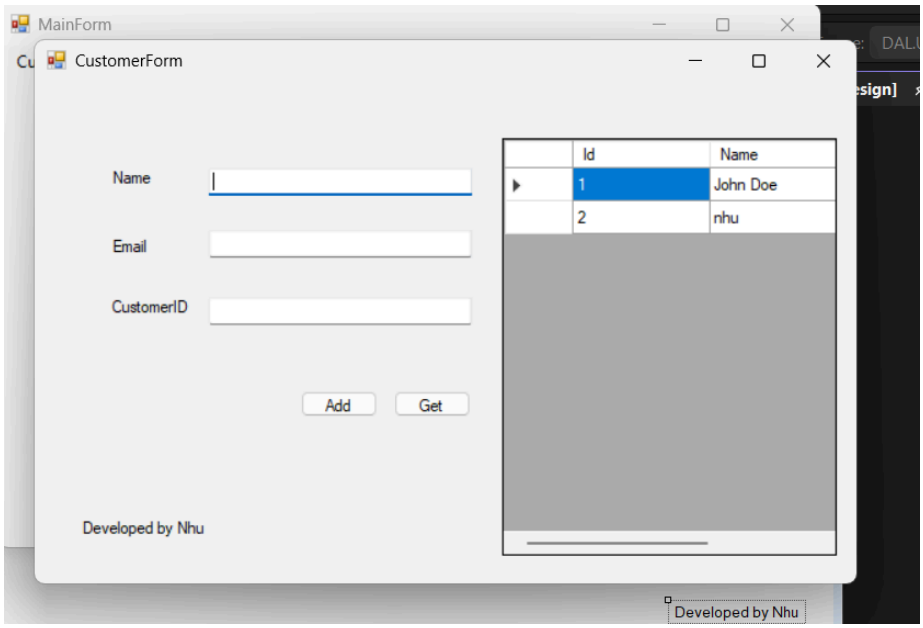
Password
Login
Cancel
Developed by Nhu
Call Stack


Debug
Any CPU
Continue
MainForm
Customer
Product
Supplier
Developed by Nhu


MainForm
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Email
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Add
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Developed by Nhu
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Name
1
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2
nhu
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